Day 03 | Monday

November 2, 2026

03

BEDMAS: choosing the correct order

12:45 PM–2:00 PM • Math • 75 min

BEDMAS in cut lists and frames

2:15 PM–3:30 PM • Math • 75 min

Week 2 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

BEDMAS: choosing the correct order

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 2, Lesson 1

2:15 PM–3:30 PM • 75 min

BEDMAS in cut lists and frames

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 2, Lesson 2

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

BEDMAS: choosing the correct order

Today’s goal: Evaluate grouped expressions and show left-to-right order for equal-ranked operations.

Name one use for today’s skill.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Exercise Book 3A Q1–10 (p.19); 3B Q2–3 (p.21)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Recall and reconnect

Compare Exercise Book 3A Q1 and Q4. Identify what changes when brackets are present.

Try first without displayed answers.

Resources: Exercise Book 3A Q1–10 (p.19); 3B Q2–3 (p.21)

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Recall from the Exercise Book | 3A Q1

$$9 + 6 \times 4$$

Write the operation, working and units before checking.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Read BEDMAS as ranked operations

1. Complete the calculation inside brackets first.
2. Evaluate exponents before multiplication or division.
3. Multiplication and division share one rank: work from left to right.
4. Addition and subtraction share one rank: work from left to right.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Make the next operation visible

1. Copy the expression exactly, including every bracket.
2. Mark the operation you will perform next.
3. Rewrite the expression after each step; carry unchanged terms forward.
4. Match the calculator’s grouping to the written expression.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Point to the next operation first

$9 + 6 \times 4 = 9 + 24 = 33$

$(9 + 6) \times 4 = 15 \times 4 = 60$

$24 \div 6 \times 2 = 4 \times 2 = 8$

The last line uses left-to-right order. Do not multiply 6 × 2 first.

Together • Exercise Book 3A Q2–7.
Point to the next operation before calculating. In Numeracy Hub, select Order of Operations → The order: BEDMAS; compare Show steps with your written working.

Textbook | page 44

Letter	Stands for	Example
B	Brackets ( ) and [ ]	(12 + 8) × 3: add 12 + 8 first
E	Exponents: squares, powers and roots	100 − 5^2: work out 5^2 first
D and M	Division and Multiplication, left to right	24 ÷ 4 × 3: divide first, because ÷ comes first
A and S	Addition and Subtraction, left to right	20 − 8 + 2: subtract first, because − comes first

Table 3.1 What each letter of BEDMAS means.

Textbook p.44 · Table 3.1: BEDMAS has four ranks, with left-to-right ties.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Textbook | page 44

Example 3.1 Bolts in a crate

A crate holds 3 loose bolts and 4 bags of 5 bolts. The count is written 3 + 4 × 5. How many bolts are in the crate?

SOLUTION

1. The 4 bags of 5 bolts are 4 × 5 = 20 bolts. That multiplication has to happen first.
2. Add the loose bolts: 3 + 20 = 23.
3. Adding first, 3 + 4 = 7 and 7 × 5 = 35, counts the 3 loose bolts as if they were in bags. That is wrong.

Textbook p.44 · Example 3.1: loose bolts plus equal groups.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Spot the mistake. Explain the correction.

In 24 ÷ 6 × 2, should you multiply 6 × 2 first because multiplication appears in BEDMAS?

Discuss the method before checking the answer.

No. Division and multiplication have equal rank. Work left to right: 24 ÷ 6 = 4, then 4 × 2 = 8.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Work together

Solve Exercise Book 3A Q2–7 with a partner. Point to the next operation before calculating.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 3A Q1–10 (p.19); 3B Q2–3 (p.21)

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Guided practice | 3A Q2

$$50 - 18 \div 3$$

Work with a partner. Take turns explaining the next step.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Compare your method | 3A Q2

$$50 - 18 \div 3$$

CHECK TOGETHER

44

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Your scheduled practice

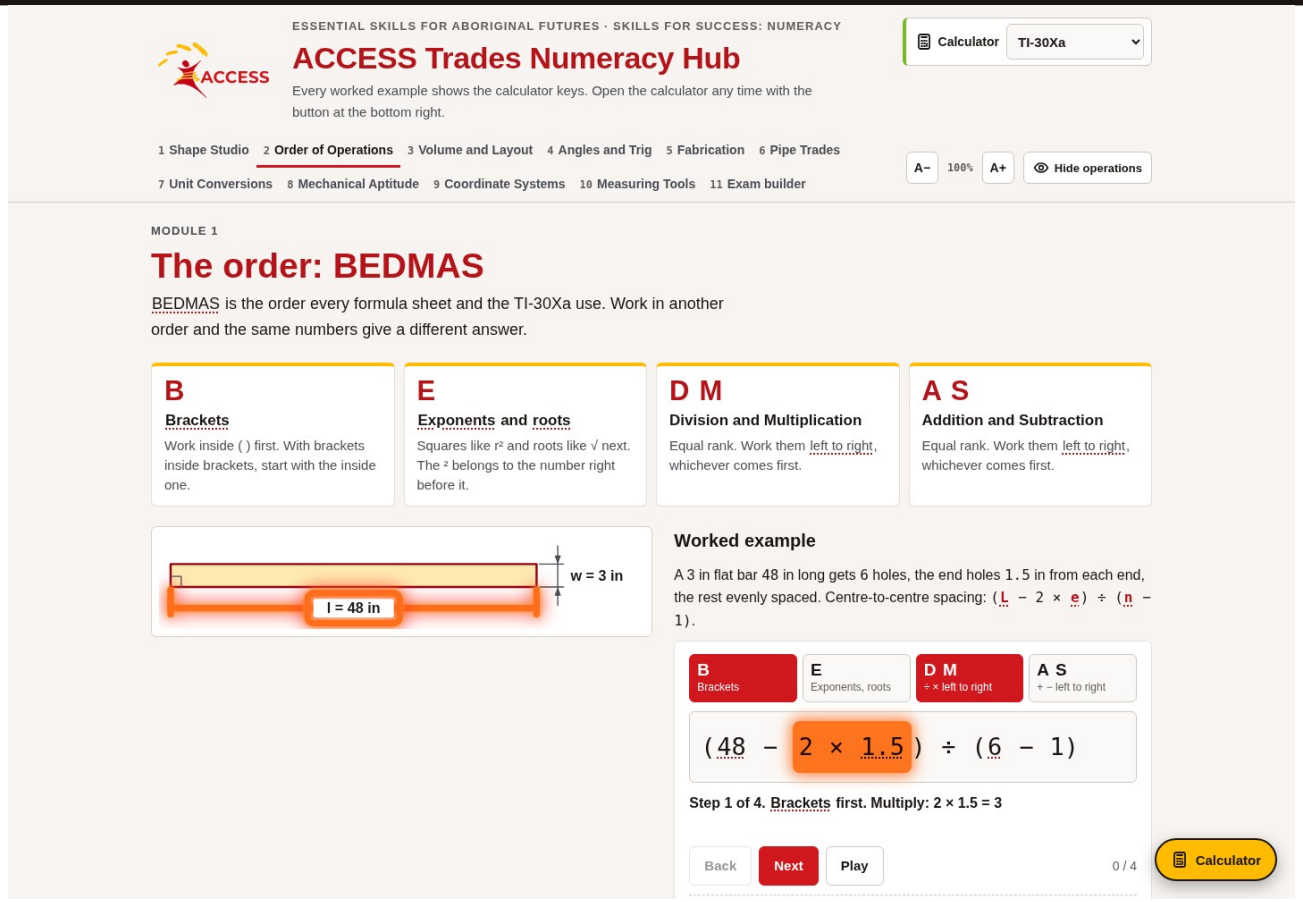
Complete Exercise Book 3A Q8–10 and 3B Q2–3. Try two Easy Numeracy Hub tasks; record expression, order and checked result, not only the screen answer.

Show working and units; estimate, then check.

Resources: Exercise Book 3A Q1–10 (p.19); 3B Q2–3 (p.21)

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Practise in Numeracy Hub

1. Supported: Open Numeracy Hub → Order of Operations → The order: BEDMAS.

2. Supported: Predict the next operation; use Show steps, Next and Back to compare with your working.

3. Independent practice: Attempt two Easy tasks on paper first; keep the expression and operation order.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Practise from the Exercise Book | 3A Q8

$6^2 - 4 \times 5$

Write the operation, working and units before checking.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Practise from the Exercise Book | 3A Q9

$$(6 - 4)^2 \times 5$$

Write the operation, working and units before checking.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Check independently

Exercise Book 3A Q8–9 without displayed solutions; mark each operation in order.

Answer independently and show the method.

Resources: Exercise Book 3A Q1–10 (p.19); 3B Q2–3 (p.21)

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Independent check | 3A Q8

$6^2 - 4 \times 5$

Answer independently. Keep worked solutions closed.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Independent check | 3A Q9

$$(6 - 4)^2 \times 5$$

Answer independently. Keep worked solutions closed.

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 3A Q1–10 (p.19); 3B Q2–3 (p.21)

Textbook Chapter 3 pp.44–51; Chapter 2 pp.31–33

Break

2:00 PM–2:15 PM

Return ready for: BEDMAS in cut lists and frames

This break is excluded from teaching minutes.

BEDMAS in cut lists and frames

Today’s goal: Write one correct expression for a cut list or frame and evaluate it in the right order.

Name one use for today’s skill.

Textbook Chapter 3 pp.44–51

Exercise Book 3B Q1–3 (p.21); 3C Q2–3 (p.23)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 3 pp.44–51

Recall and reconnect

Compare 2 × (10 + 5) with 2 × 10 + 5. Try to explain the grouping.

Try first without displayed answers.

Resources: Exercise Book 3B Q1–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 3 pp.44–51

Make brackets represent the drawing

1. Repeated pieces become number of pieces × length per piece.
2. Group every quantity that must be removed from the original stock.
3. Count only the cuts and kerf explicitly stated in the problem.
4. Offcut = stock − (all piece lengths + all stated cutting loss).

Textbook Chapter 3 pp.44–51

Frames count pairs of sides

1. A rectangle has two lengths and two widths.
2. Adding four sides and calculating 2 × (length + width) count the same boundary.
3. In 2 × (length + width), the 2 multiplies both dimensions.
4. Keep the sketch, expression and measurement unit attached to the same problem.

Textbook Chapter 3 pp.44–51

Subtract the entire amount used

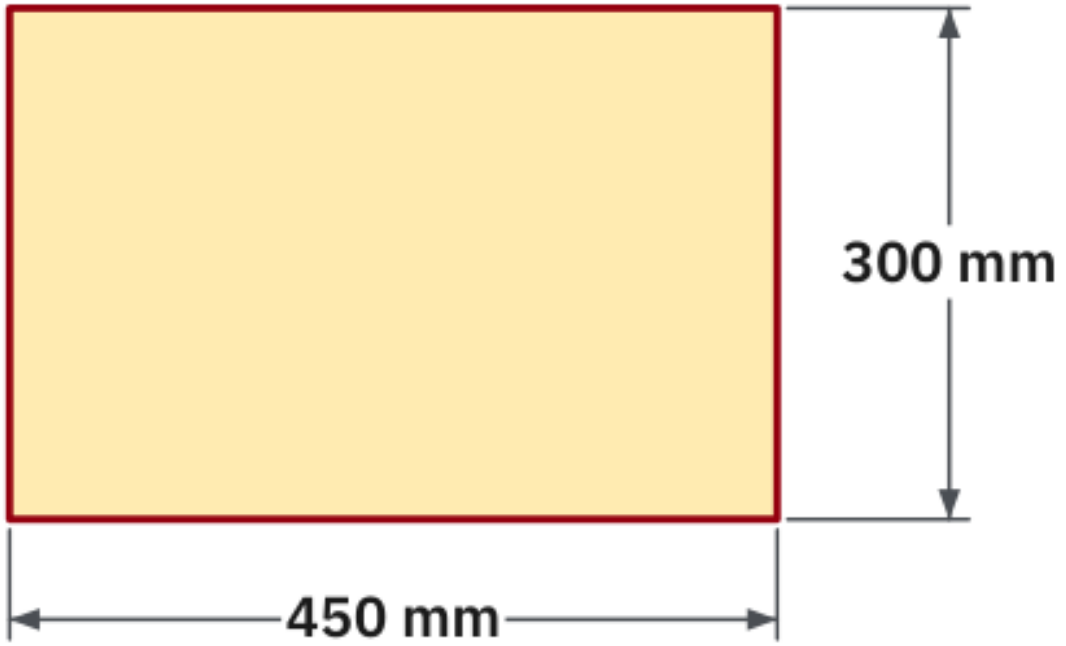
Classroom example: 3,000 mm stock; two 850 mm pieces;
two stated cuts, each with 3 mm kerf.

Offcut = 3,000 − (2 × 850 + 2 × 3)
= 3,000 − 1,706 = 1,294 mm

Check: 1,700 mm pieces + 6 mm kerf + 1,294 mm offcut = 3,000 mm.

Together • Exercise Book 3B Q1–2.
Underline the group before entering keys. Draw the pieces and label what each number counts.

Textbook | page 46



Textbook p.46 · Frame diagram: two lengths and two widths.

Textbook Chapter 3 pp.44–51

Textbook | page 46

Example 3.3 Brackets change the answer

Find 12 + 8 × 3 and (12 + 8) × 3.

SOLUTION

1. 12 + 8 × 3: multiply first, 8 × 3 = 24, then 12 + 24 = 36.
2. (12 + 8) × 3: brackets first, 12 + 8 = 20, then 20 × 3 = 60.

Textbook p.46 · Example 3.3: brackets change what is multiplied.

Textbook Chapter 3 pp.44–51

Spot the mistake. Explain the correction.

Does stock − pieces + kerf subtract the cutting loss?

Discuss the method before checking the answer.

No. It adds kerf back. Subtract the full group: stock − (pieces + kerf).

Textbook Chapter 3 pp.44–51

Work together

Solve 3B questions 1–2. Underline the group inside brackets before entering keys.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 3B Q1–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 3 pp.44–51

Guided practice | 3B Q1

A spool piece has 6 flanges with 8 bolts each. You also want 4 spare bolts. Write one calculation for the bolts needed and work it out.

Work with a partner. Take turns explaining the next step.

Textbook Chapter 3 pp.44–51

Compare your method | 3B Q1

A spool piece has 6 flanges with 8 bolts each. You also want 4 spare bolts. Write one calculation for the bolts needed and work it out.

CHECK TOGETHER

6 × 8 + 4 = 52 bolts

Textbook Chapter 3 pp.44–51

Your scheduled practice

Complete 3B question 3 and 3C questions 2–3. Draw the pieces, write one expression and compare with the Numeracy Hub steps.

Show working and units; estimate, then check.

Resources: Exercise Book 3B Q1–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 3 pp.44–51

Open Numeracy Hub

ESSENTIAL SKILLS FOR ABORIGINAL FUTURES · SKILLS FOR SUCCESS: NUMERACY

ACCESS Trades Numeracy Hub

Every worked example shows the calculator keys. Open the calculator any time with the button at the bottom right.

Calculator TI-30Xa

1 Shape Studio | 2 Order of Operations | 3 Volume and Layout | 4 Angles and Trig | 5 Fabrication | 6 Pipe Trades | 7 Unit Conversions | 8 Mechanical Aptitude | 9 Coordinate Systems | 10 Measuring Tools | 11 Exam builder

A– 100% A+ Hide operations

MODULE 1

The order: BEDMAS

BEDMAS is the order every formula sheet and the TI-30Xa use. Work in another order and the same numbers give a different answer.

B — Brackets
Work inside ( ) first. With brackets inside brackets, start with the inside one.

E — Exponents and roots
Squares like r² and roots like √ next. The ² belongs to the number right before it.

D M — Division and Multiplication
Equal rank. Work them left to right, whichever comes first.

A S — Addition and Subtraction
Equal rank. Work them left to right, whichever comes first.



Worked example

A 3 in flat bar 48 in long gets 6 holes, the end holes 1.5 in from each end, the rest evenly spaced. Centre-to-centre spacing: (L − 2 × e) ÷ (n − 1).

B Brackets | E Exponents, roots | D M ÷ × left to right | A S + − left to right

(48 − 2 × 1.5) ÷ (6 − 1)

Step 1 of 4. Brackets first. Multiply: 2 × 1.5 = 3

Back | Next | Play | 0 / 4

Calculator

Navigation example. Use the values and units in the assigned task.

Textbook Chapter 3 pp.44–51

Practise in Numeracy Hub

1. Supported: Draw and label the pieces or frame for the assigned question; write one expression.

2. Independent practice: Complete the paper calculation and underline the grouped quantity.

3. Independent checking: Open Numeracy Hub BEDMAS and compare Show steps with the written order.

Textbook Chapter 3 pp.44–51

Practise from the Exercise Book | 3B Q3

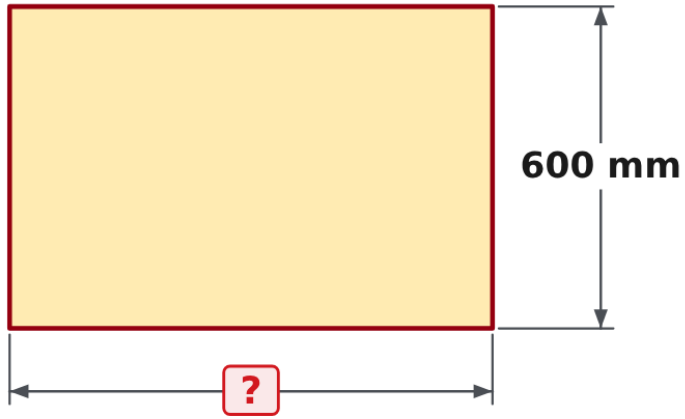
Four 27 in pieces and two 15 in pieces are cut from a 144 in bar. Ignore the saw cuts. The offcut is 144 − (4 × 27 + 2 × 15). Find it.

Write the operation, working and units before checking.

Textbook Chapter 3 pp.44–51

Practise from the Exercise Book | 3C Q2

The distance around this frame is 3,000 mm. Find the hidden length with $L = 3{,}000 \div 2 - 600$.



Write the operation, working and units before checking.

Textbook Chapter 3 pp.44–51

Check independently

Redo 3C questions 2–3 and explain the brackets.

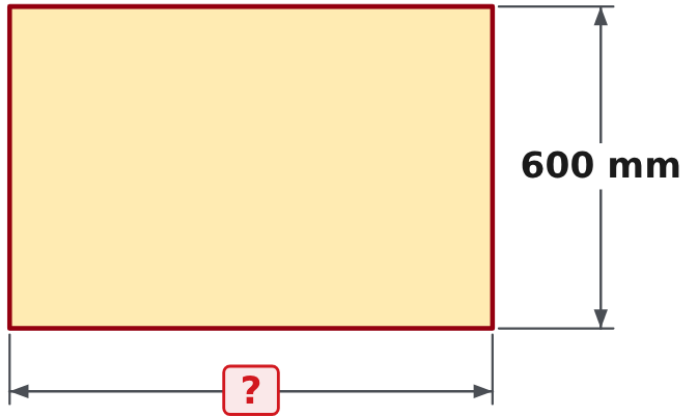
Answer independently and show the method.

Resources: Exercise Book 3B Q1–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 3 pp.44–51

Independent check | 3C Q2

The distance around this frame is 3,000 mm. Find the hidden length with $L = 3{,}000 \div 2 - 600$.

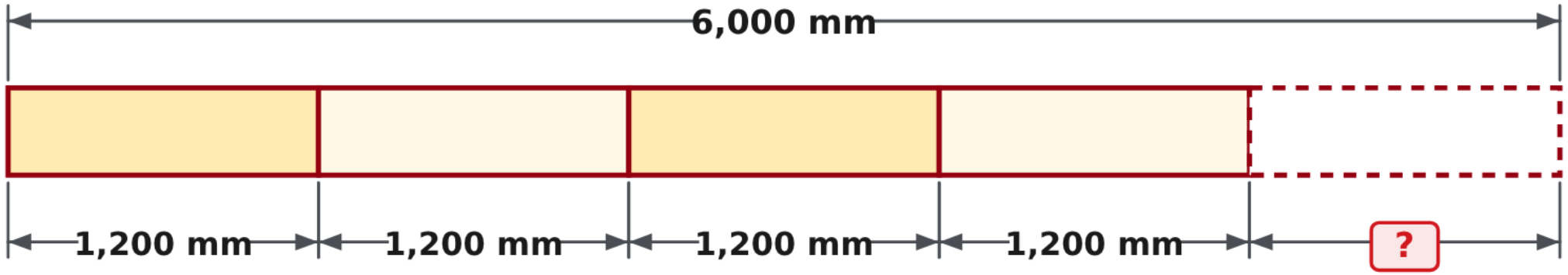


Answer independently. Keep worked solutions closed.

Textbook Chapter 3 pp.44–51

Independent check | 3C Q3

Four 1,200 mm pieces are cut from a 6,000 mm bar with four cuts. The kerf is 3 mm. The offcut is 6,000 − (4 × 1,200 + 4 × 3). Find it.



Answer independently. Keep worked solutions closed.

Textbook Chapter 3 pp.44–51

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 3B Q1–3 (p.21); 3C Q2–3 (p.23)

Textbook Chapter 3 pp.44–51